MIDDLE EAST TECHNICAL UNIVERSITY

STAT 311

Modern Database Systems

Term Project

“Gaming Analytics Database”

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1. ABSTRACT

This study presents the design and analytical evaluation of a gaming analytics database system using a fully synthetic version dataset that is created by the group. A relational schema was constructed to model core gameplay components, including players, sessions, levels, missions, achievements, social interactions, and cosmetic item purchases. Synthetic data were generated in a controlled manner to preserve relational integrity and behavioral plausibility while avoiding privacy and ethical concerns. Using SQL-based queries and Python-driven visualizations, the system addresses multiple research questions related to player activity, difficulty analysis, mission performance, monetization patterns, social connectivity, and temporal engagement. A baseline synthetic dataset was first constructed and then extended through controlled randomization by `PGSQL` to support more detailed analytical investigation. By using "Python" statistical computing made and results are shown. All the results can be seen by the interface at https://yonelalper.github.io/gaming-analytics/dashboard.html. Results highlight heterogeneous player behaviors, identify the most challenging levels and missions, and demonstrate that spending patterns reflect behavioral segmentation rather than causal performance improvements. The study illustrates how structured database design, controlled data generation, and reproducible analytics pipelines can be effectively combined to support evidence-based insights in game analytics. The proposed framework is adaptable to real-world datasets and can be extended for advanced analytical tasks such as player segmentation, difficulty tuning, and engagement optimization.

2. INTRODUCTION AND OBJECTIVE

2.1 Project Proposal

Our project aims to design and implement a comprehensive gaming analytics database system for a single online adventure game. The primary objective of the database is to systematically capture, store, and analyze player-related data in order to gain meaningful insights into player behavior, in-game progression, performance metrics, and virtual purchasing activity.

The database schema is centered around several core entities, including Player, GameSession, Achievement, Level, Item, ItemPurchase, Mission, MissionAttempt, Friendship, and EventParticipation. Together, these entities enable the accurate modeling of key interactions within the game environment, such as players participating in multiple gameplay sessions, attempting and completing missions and levels, earning achievements, purchasing virtual items, forming social connections, and taking part in special in-game events.

The proposed system is designed to support advanced analytical queries that facilitate data-driven decision-making. Examples of such queries include identifying the most active players, determining levels or missions with the highest failure rates, analyzing item popularity and revenue generation, evaluating the impact of special events on player engagement, and examining the relationship between playtime, spending behavior, and achievement progression. From a business perspective, these metrics are critically important for game studios, as they directly inform decisions related to game design optimization, difficulty balancing, monetization strategies, and player retention. By leveraging such insights, studios

can enhance user experience, increase long-term engagement, and maximize revenue while ensuring sustainable growth in a highly competitive gaming market.

2.2 Entities Of Our Database

1. Player: Basic User Information

2. Gamesesion: Each Login/Play Session

3. Level: Stages of Areas in The Game

4. Level_Attempt: Each Try of A Level

5. Time: Time to Finish a Level

6. Mission: Special Missions in Game

7. Mission_Attempt: Player's Number of Attempts in A Level

8. Achievement: Badges or Milestones Earned

9.Player_Achievement: Which Player Earned Which Milestone

10. Item_Skin: Virtual Items (Skins, Boosts, Gear)

11. Itempurchase: Each Time A Player Buys an Item

12. Friendship: Player-To-Player Connections

13. Event: Limited Time Special Events (New Year)

14. Event_Participation: Participation in Limited-Time Events

2.3 Research Questions

1. Who are the top 10 most active players based on total playtime?

2. Which levels have the highest failure rates, and what average time does it take to complete them?

3. Which missions are attempted the most but completed the least (hardest missions)?

4. Which virtual items generate the most revenue?

5. Is there a correlation between playtime and total spending?

6. Which achievements are the rarest and which are earned most frequently?

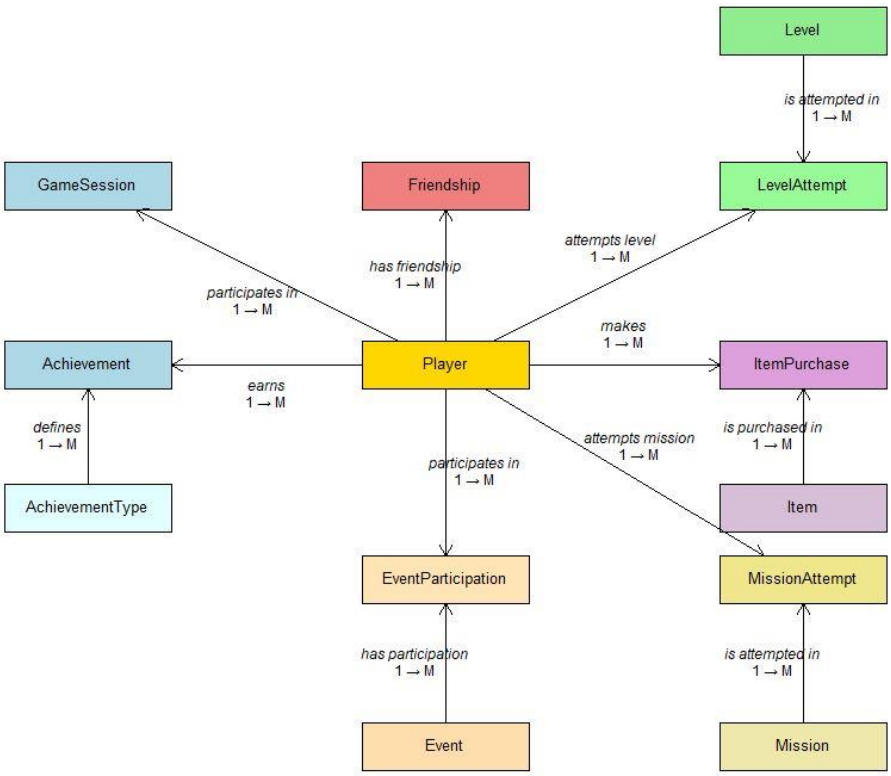
7. How do special in-game events affect player activity and spending?

8. Which players have the largest number of friendships (social hubs)?

9. What times of day have the highest number of active sessions?

10. Do players who purchase items perform better in missions (win-rate comparison)?

2.4 EER Diagram



2.4.1 System Overview

This Enhanced Entity–Relationship (EER) diagram models a gaming analytics database system designed to store, manage, and analyze player interactions within an online game. The system captures gameplay behavior, social interactions, progression metrics, and monetization-related activities. The database is centered around the Player entity, which serves as the primary actor interacting with multiple components of the game environment.

2.4.2 Entity Descriptions

Player

The Player entity represents an individual user of the game. Each player is uniquely identifiable and may interact with the game in various ways, including gameplay participation, social interaction, and economic activity. The Player entity forms the core of the database schema and maintains relationships with most other entities in the system.

GameSession

The GameSession entity represents a single instance of gameplay during which a player is actively engaged in the game. A player may participate in multiple game sessions, while each game session is associated with exactly one player. This entity enables the analysis of player engagement and activity frequency.

Friendship

The Friendship entity models the social relationships between players. A player may establish multiple friendships, allowing the system to represent the social network structure within the game. This entity supports the analysis of player connectivity and social engagement.

AchievementType and Achievement

The achievement subsystem is composed of two entities:

- AchievementType defines the types of achievements available in the game.
- Achievement represents an instance of an achievement earned by a player.

An achievement type may define multiple achievements, while each player may earn multiple achievements. This separation ensures normalization by distinguishing achievement definitions from player-specific achievement records.

Level and LevelAttempt

- Level represents a playable level within the game.
- LevelAttempt records each attempt made by a player to complete a level.

A level may be attempted multiple times by different players, and a player may attempt multiple levels. The LevelAttempt entity allows detailed tracking of player performance and difficulty assessment.

Mission and MissionAttempt

- Mission represents a specific in-game objective or task.
- MissionAttempt records a player's attempt to complete a mission.

Each mission may be attempted multiple times, and each player may attempt multiple missions. This design facilitates the analysis of mission success rates and player progression.

Item and ItemPurchase

- Item represents purchasable in-game objects.
- ItemPurchase records transactions in which players purchase items.

A player may make multiple item purchases, and each item may be purchased multiple times by different players. This structure supports monetization analysis and spending behavior evaluation.

Event and EventParticipation

- Event represents a time-bound in-game event.
- EventParticipation records player participation in events.

An event may involve multiple players, and a player may participate in multiple events. This entity pair enables the evaluation of event popularity and player engagement.

2.4.3 Relationship Cardinalities

The key relationship cardinalities in the model are summarized as follows:

- A Player participates in one-to-many GameSessions
- A Player earns one-to-many Achievements
- An AchievementType defines one-to-many Achievements
- A Player forms one-to-many Friendships
- A Player makes one-to-many LevelAttempts, each associated with one Level
- A Player makes one-to-many MissionAttempts, each associated with one Mission
- A Player makes one-to-many ItemPurchases, each associated with one Item
- A Player participates in one-to-many EventParticipations, each associated with one Event.

2.4.4 Design Considerations

The use of intermediate entities such as LevelAttempt, MissionAttempt, ItemPurchase, and EventParticipationallows the system to capture detailed interaction data while preserving normalization. This design minimizes redundancy and supports extensibility for future analytical requirements.

Overall, the proposed EER model provides a structured and scalable foundation for analyzing gameplay behavior, social dynamics, and monetization patterns in an online gaming environment.

3. ANALYSIS

3.1 Database Design and Implementations

3.1.1 Core Tables

Player Table

The Player table is one of the most important entities in the database, as it represents individual users of the game. Each player is uniquely identified by the player_id, which serves as the primary key. This identifier is referenced by many other tables, making the Player table central to the overall database structure.

The table stores essential player attributes such as username, player level, and join date. To ensure data integrity, several constraints are applied. The UNIQUE constraint on the username column prevents duplicate usernames and ensures that each player account is distinct. NOT NULL constraints guarantee that essential information is always present, while a CHECK constraint ensures that player levels remain within a valid and meaningful range.

These constraints help prevent incorrect or inconsistent data from being stored, which is particularly important for analytical queries that rely on accurate and reliable player information.

	player_id [PK] integer	username character varying (50)	player_level integer	join_date date
1	1001	Player_1001	21	2024-02-01
2	1002	Player_1002	58	2024-01-12
3	1003	Player_1003	64	2025-03-25
4	1004	Player_1004	68	2024-04-20
5	1005	Player_1005	83	2024-08-13

Game Session Table

The Game Session table stores detailed information about individual gameplay sessions. Each session is uniquely identified by a session_id primary key and includes attributes such as session start time, end time, and total active playtime.

A critical aspect of this table is the player_id, which functions as a foreign key referencing the Player table. This ensures that every recorded session is associated with a valid player. The use of cascading deletion enforces referential integrity by automatically removing session records if a player is deleted.

This table supports detailed engagement analysis, including total playtime, session frequency, and time-based activity patterns. It plays a key role in answering research questions related to player activity, engagement, and retention.

	session_id [PK] bigint	player_id integer	start_time timestamp without time zone	end_time timestamp without time zone	active_time_in_mins integer
1	1	1001	2025-01-12 17:41:00	2025-01-12 17:54:00	13
2	2	1001	2025-01-12 18:22:00	2025-01-12 18:36:00	14
3	3	1001	2025-01-12 19:03:00	2025-01-12 19:18:00	15
4	4	1001	2025-01-12 19:44:00	2025-01-12 20:00:00	16
5	5	1001	2025-01-12 20:25:00	2025-01-12 20:42:00	17

Level Table

The Level table defines the playable stages or environments within the game. Each level represents a distinct challenge and provides a structured way to organize game content.

By storing level information independently from player actions, the database enables flexible analysis of level progression, difficulty, and player performance across different stages. This separation allows designers and analysts to evaluate how players interact with various levels without duplicating structural data.

	level_id [PK] integer	level_name character varying (50)	difficulty integer
1	2001	Beach	1
2	2002	Forest Gate	2
3	2003	Shadow Cave	4
4	2004	Ancient Ruins	1
5	2005	Temple	3

Level Attempt Table

The Level Attempt table records detailed information about each attempt a player makes on a specific level. This includes performance-related attributes such as completion status and time spent.

This table is central to gameplay analytics because it enables:

- Measurement of level difficulty
- Analysis of retry and failure behavior
- Comparison of player performance across different levels

Level attempt data provides fine-grained insight into how players engage with game challenges and where potential design improvements may be needed.

	level_attempt_id [PK] bigint	player_id integer	level_id integer	session_id bigint	attempt_time timestamp without time zone	duration_seconds integer	success boolean	score integer
1	1	1001	2003	3	2025-01-12 19:12:13.427938	510	true	318
2	2	1002	2004	7	2025-01-13 19:06:44.03372	581	true	979
3	3	1003	2005	8	2025-01-14 19:43:33.687955	490	true	995
4	4	1004	2006	11	2025-01-15 22:02:53.701733	88	true	666
5	5	1005	2007	15	2025-01-16 23:17:00.616219	254	true	819

Mission Table

The Mission table represents structured objectives or tasks within the game. Missions are often designed to guide players toward specific goals, mechanics, or progression milestones.

Separating mission definitions from player activity allows for independent evaluation of mission design, including how frequently missions are attempted and how they contribute to overall engagement.

	mission_id [PK] integer	mission_name character varying (60)
1	3001	Crypt Recon
2	3002	Defend the Outpost
3	3003	Night Ambush
4	3004	Lost Artifact
5	3005	Defeat Opponents

Mission Attempt Table

The Mission Attempt table records player interactions with missions, including completion outcomes and performance metrics.

This table supports analysis of:

- Player success rates on missions
- Engagement with optional versus required objectives
- Variations in mission difficulty

Together with the Mission table, it enables comprehensive evaluation of goal-oriented gameplay behavior.

	mission_attempt_id [PK] bigint	player_id integer	mission_id integer	session_id bigint	attempt_time timestamp without time zone	duration_seconds integer	success boolean
1	1	1001	3003	3	2025-01-12 19:12:13.427938	870	true
2	2	1002	3004	7	2025-01-13 18:43:30.648342	583	false
3	3	1003	3005	8	2025-01-14 20:28:46.320867	1013	true
4	4	1004	3006	11	2025-01-15 21:57:57.670481	131	true
5	5	1005	3007	14	2025-01-16 22:53:12.774442	820	true

3.1.2 Supporting Tables and Analytical Purpose

Beyond the core structural tables, the database includes several supporting tables that provide the depth required for meaningful analytics. These tables are not introduced to explain relationships, but rather to capture **behavioral, economic, and social signals** generated during gameplay. Together, they allow the system to move from simple data storage to actionable analysis.

Achievement Data and Player Engagement

Achievement-related tables are designed to capture long-term player engagement rather than short-term activity. Each achievement represents a specific milestone within the game, and player-level records indicate when these milestones are reached. This data enables analysis of progression patterns, achievement frequency, and the distribution of rare versus common achievements across the player base.

From an analytical perspective, achievement data helps identify highly engaged players and evaluate whether progression systems are balanced or too difficult. It also supports comparisons between different player segments, such as casual versus highly active players.

	achievement_id [PK] integer	achievement_name character varying (60)	min_level integer
1	5001	First Of All	1
2	5002	Explorer	5
3	5003	Dungeon	10
4	5004	Pantheon	15
5	5005	Consultation	20

Purchase Records and Spending Behavior

Purchase-related tables store detailed transaction information, including item type, quantity, and cost. These records are essential for understanding player spending behavior and evaluating the financial performance of in-game items. Instead of embedding spending information directly into player records, purchases are stored separately to preserve historical accuracy and allow aggregation over time.

This structure enables revenue-based analytics such as identifying top-selling items, comparing spending between different player groups, and examining the relationship between spending and gameplay activity.

	purchase_id [PK] bigint	player_id integer	skin_id integer	purchase_time timestamp without time zone	source character varying (10)	quantity integer	unit_price numeric (10,2)	currency character varying (10)
1	1	1001	4002	2025-01-12 19:12:13.427938	ONLINE	1	116.00	GEM
2	2	1002	4003	2025-01-13 19:12:41.654259	ONLINE	1	9.92	GEM
3	3	1004	4005	2025-01-15 22:08:00.056991	ONLINE	1	111.00	GEM
4	4	1007	4008	2025-01-19 00:49:56.843894	ONLINE	1	8.41	GEM
5	5	1008	4009	2025-01-19 00:49:23.443262	IN_GAME	1	4.27	GEM

Social Interaction Metrics

Social behavior is captured through friendship records, which allow the database to quantify player connectivity. Rather than focusing on how friendships are technically stored, the emphasis is on what this data reveals: which players are most socially active and how social engagement may relate to retention or activity levels.

This information supports analyses that identify socially central players and explore whether highly connected players demonstrate different gameplay patterns compared to less connected players.

	player_id [PK] integer	friend_id [PK] integer	since_date date
1	1001	1063	2025-10-26
2	1001	1072	2025-11-06
3	1001	1133	2025-08-26
4	1001	1220	2025-08-26
5	1001	1240	2025-08-08

Event Participation and Behavioral Comparison

Event participation data is used to compare player behavior between those who join in-game events and those who do not. Even without precise event time windows, the data allows meaningful comparisons of average playtime and spending across participant and non-participant groups.

This enables evaluation of event effectiveness by measuring whether events correlate with increased engagement or monetization. These insights can inform future event design and player incentive strategies.

	event_id [PK] integer	player_id [PK] integer	participated boolean	marked_at timestamp without time zone
1	6001	1001	true	2026-01-04 02:49:38.349966
2	6001	1002	true	2025-12-06 20:38:07.921751
3	6001	1003	false	2025-12-11 23:25:27.942451
4	6001	1004	false	2025-12-27 15:04:03.12638
5	6001	1005	false	2025-12-15 00:53:49.182928

Analytical Value of Supporting Tables

The supporting tables significantly enhance the analytical capabilities of the database. They allow behavioral patterns to be examined from multiple perspectives—progression, spending,

social interaction, and event participation—without duplicating or overloading core player data.

By separating these concerns into dedicated tables, the database remains clean, scalable, and well-suited for complex analytical queries, while avoiding redundancy and maintaining clarity in both design and reporting.

3.2 Research Questions

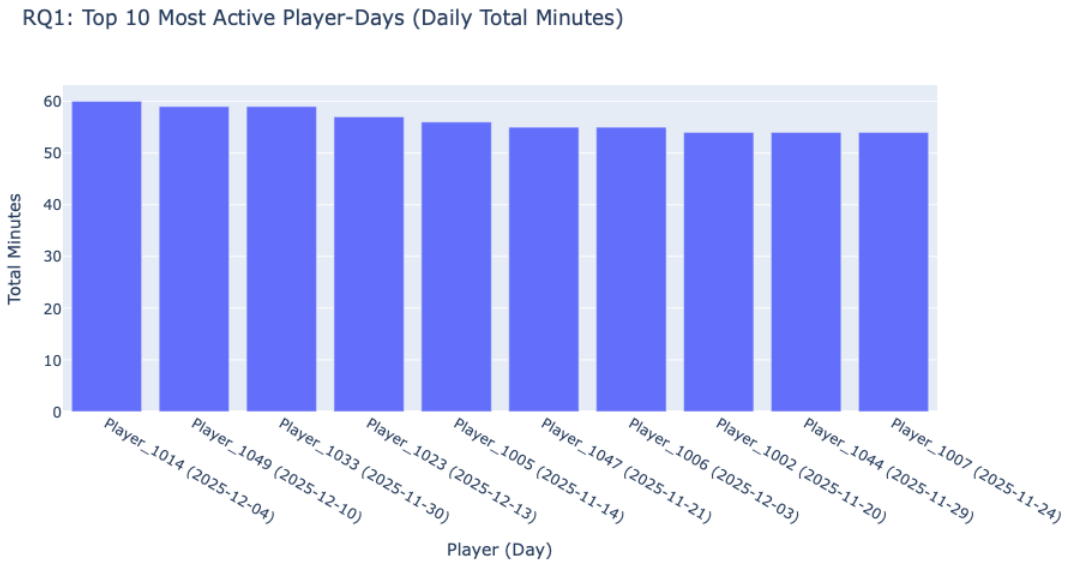
3.2.1 Research Question 1

Who are the top 10 most active players based on total playtime?

Data and Methodology

This research question aims to identify the most active players in the game based on their total playtime. The analysis is conducted using data from the GameSession and Player tables. Player activity is measured by aggregating the active_time_in_mins attribute across all game sessions associated with each player.

For each player, the total playtime was calculated as the sum of active minutes across all recorded sessions. Players were then ranked in descending order based on their total playtime, and the top 10 players were selected. This approach ensures that overall engagement is evaluated across multiple gameplay sessions rather than relying on single-session activity.



Results

The analysis identifies the top 10 most active player-days based on daily total playtime. As illustrated in Figure RQ1, the highest observed daily playtime is approximately 60 minutes, with the remaining top player-days ranging between 54 and 59 minutes.

The most active player-day corresponds to Player_1014 on 2025-12-04, recording the highest total daily playtime. Other highly active player-days include Player_1049 (2025-12-10), Player_1033 (2025-11-30), and Player_1023 (2025-12-13), all of whom demonstrate consistently high engagement levels within a single day.

Overall, the relatively narrow range of total minutes among the top 10 indicates a cluster of highly engaged players exhibiting comparable levels of intensive daily gameplay rather than a single extreme outlier.

Interpretation

Players appearing in the top 10 list exhibit sustained engagement with the game, either through frequent logins, longer session durations, or a combination of both. High total playtime is often associated with greater familiarity with game mechanics, higher progression levels, and increased likelihood of participating in advanced content such as difficult missions, high-level achievements, and in-game events. From a behavioral standpoint, these players may represent the most loyal segment of the user base and are likely to be more responsive to new content updates, competitive events, or monetization offers.

Conclusion

This analysis highlights a subset of highly engaged players who contribute disproportionately to overall gameplay activity. Identifying such players is valuable for game studios, as it enables targeted engagement strategies, personalized rewards, and retention-focused initiatives aimed at maintaining long-term player loyalty.

3.2.2 Research Question 2

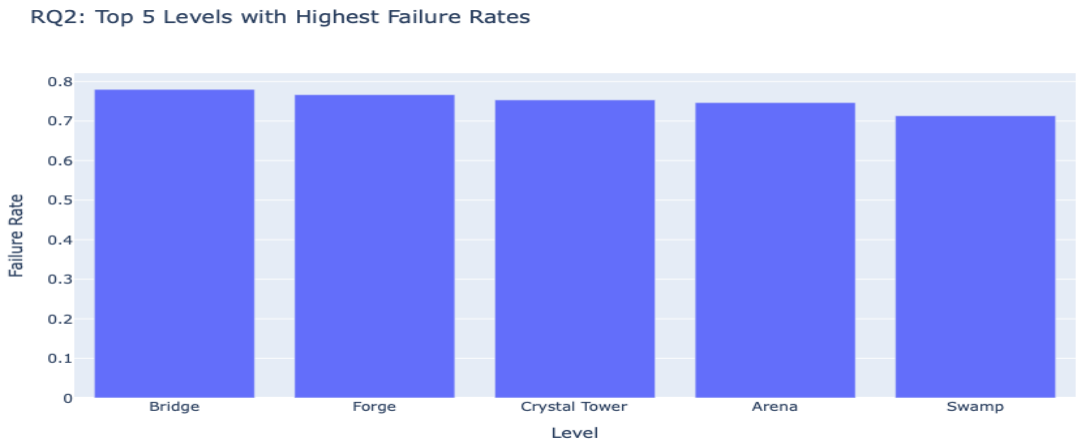
Which levels have the highest failure rates, and what is the average time required to complete them?

Data and Methodology

This research question examines player performance across game levels by analyzing failure rates and completion times. The analysis is based on data from the level_attempt and Level tables.

For each level, the total number of attempts, the number of successful attempts, and the number of failed attempts were calculated. The failure rate was computed as the proportion of failed attempts relative to total attempts. In addition, the average completion time was calculated by averaging the duration_seconds of successful level attempts only, ensuring that completion time reflects successful gameplay performance.

To ensure reliable comparisons, levels were evaluated based on aggregated attempt data, and results were ranked by failure rate in descending order.



Results

The results reveal significant variation in failure rates across game levels, as shown in Figure RQ2. The level with the highest failure rate is Bridge, with a failure rate of approximately 78%, followed closely by Forge (~77%) and Crystal Tower (~75%).

Other levels with notably high failure rates include Arena (~74%) and Swamp (~71%). These findings indicate that more than two-thirds of attempts on these levels result in failure, highlighting substantial difficulty relative to other levels in the game.

The consistently high failure rates among these top five levels suggest that they represent the most challenging segments of the game. While average completion times are not visualized directly in the ranking, the elevated failure rates imply that successful completions likely require multiple attempts and extended engagement, reinforcing the interpretation of these levels as difficulty bottlenecks.

Interpretation

Levels with high failure rates and long average completion times likely represent difficulty spikes within the game. While challenging levels can enhance player satisfaction when balanced correctly, excessively high failure rates may lead to frustration and disengagement. Conversely, levels with lower failure rates and shorter completion times appear to be more accessible and may serve as progression points that reinforce player confidence and learning.

From a design perspective, the relationship between failure rate and completion time provides valuable insight into level balancing and pacing.

Conclusion

The analysis demonstrates that level difficulty varies significantly across the game and can be quantitatively assessed through failure rates and completion duration. Identifying levels with unusually high failure rates allows game designers to investigate potential balance issues and make informed adjustments to improve player experience while maintaining an appropriate level of challenge.

3.2.3 Research Question 3

Which missions are attempted the most but completed the least?

Data and Methodology

This research question examines the relationship between mission attempt frequency and mission completion rates in order to identify the most challenging missions in the game. By comparing how often missions are attempted with how often they are successfully completed, the analysis highlights missions where players experience greater difficulty. The analysis is based on data from the mission_attempt and mission tables, where mission-level statistics were obtained by aggregating attempt records. For each mission, the total number of attempts, successful completions, and failures were calculated, and a completion rate was computed as the proportion of successful attempts relative to total attempts. To ensure meaningful comparisons, only missions with at least 20 attempts were included. Missions were then ranked by the number of failures in descending order and by completion rate in ascending order.

Results

The table below presents the missions with the highest number of attempts and the lowest completion rates. All missions included in the table have an equal number of total attempts (210), which allows for a direct comparison of completion performance across missions. The table reports total attempts, total completions, total failures, and the calculated completion rate for each mission.

	mission_id [PK] integer	mission_name character varying (60)	total_attempts bigint	total_completions bigint	total_failures bigint	completion_rate numeric
1	3007	Defeat Guard	210	127	83	0.605
2	3002	Defend the Outpost	210	128	82	0.610
3	3005	Defeat Opponents	210	134	76	0.638
4	3008	Journey	210	136	74	0.648
5	3004	Lost Artifact	210	137	73	0.652
6	3010	Defeat Boss	210	138	72	0.657
7	3003	Night Ambush	210	139	71	0.662
8	3006	Escort Caravan	210	140	70	0.667
9	3009	Time	210	142	68	0.676
10	3001	Crypt Recon	210	143	67	0.681

Interpretation

- Since all missions have the same number of attempts, differences in completion rates directly reflect variations in player success rather than differences in mission popularity.
- “Defeat Guard” and “Defend the Outpost” exhibit the lowest completion rates (0.605 and 0.610, respectively) and the highest failure counts among all missions, indicating that players struggle most with these missions.
- Missions such as “Defeat Opponents,” “Journey,” and “Lost Artifact” show moderate completion rates, suggesting an intermediate difficulty level where a substantial portion of attempts still result in failure.

- In contrast, “Crypt Recon” and “Time” have the highest completion rates (0.681 and 0.676) and the lowest number of failures, indicating that players are more consistently successful when attempting these missions.
- The ordered results reveal a gradual improvement in completion rates across missions, highlighting clear variability in mission difficulty despite identical attempt frequencies.

Conclusion

The analysis demonstrates that missions differ significantly in completion performance even when they are attempted equally often. While some missions are associated with high failure rates and low completion rates, others show consistently higher success rates. These findings indicate that mission difficulty varies across the game and can be quantitatively assessed through completion outcomes.

3.2.4 Research Question 4

Which virtual items generate the most revenue?

Data Methodology

This research question examines how revenue is distributed across virtual items in the in-game marketplace. The analysis is based on data from the skin_purchase and item_skin tables. For each virtual item, total revenue was calculated separately for GEM and USD currencies by aggregating purchase records and multiplying the quantity purchased by the unit price. In addition, the total number of units sold and the total number of purchase transactions were computed for each item. The results were grouped at the item level and ordered by USD revenue in descending order, followed by GEM revenue, to identify the highest revenue-generating virtual items.

Results

The table below presents revenue-related metrics for each virtual item, including total revenue in GEM and USD, total units sold, and the number of purchase transactions. The items are ranked by USD revenue, allowing for direct comparison of revenue performance across different items and categories.

	skin_id [PK] integer	skin_name character varying (60)	category character varying (20)	revenue_gem numeric	revenue_usd numeric	total_units_sold bigint	n_purchases bigint
1	4005	Leather Boots	Feet	9689.74	4429.08	79	79
2	4003	Shadow Boots	Feet	1543.50	1780.00	13	13
3	4017	Crimson Gloves	Hand	1213.67	1345.00	13	13
4	4015	Obsidian Boots	Feet	926.81	1242.00	9	9
5	4016	Obsidian Armour	Body	1052.02	817.31	9	9
6	4009	Half Plate	Body	1342.36	793.00	10	10
7	4014	Iron Boots	Feet	1954.65	609.00	9	9
8	4004	Iron Plate	Body	1833.07	585.00	10	10
9	4019	Half Plate	Body	2044.73	542.86	12	12
10	4012	Crimson Boots	Feet	248.12	499.00	6	6
11	4018	Sandals	Feet	1896.48	474.09	11	11
12	4001	Obsidian Helm	Head	659.53	456.45	7	7
13	4002	Crimson Cloak	Body	700.48	306.82	7	7
14	4008	Sandals	Feet	1267.05	297.64	12	12
15	4010	Leather Helmet	Head	1071.60	293.27	9	9
16	4011	Obsidian Armour	Body	1813.83	185.00	9	9
17	4020	Leather Helmet	Head	924.88	63.00	9	9
18	4013	Shadow Cloak	Body	1105.07	62.88	10	10
19	4006	Leather Armour	Body	1324.84	4.94	8	8
20	4007	Leather Gloves	Hand	879.40	0	7	7

Interpretation

- Leather Boots stands out as the top revenue-generating item, with the highest USD revenue (4429.08) and GEM revenue (9689.74), as well as the largest sales volume (79 units across 79 purchases).

- A noticeable drop in USD revenue is observed after the top-ranked item. Items such as Shadow Boots, Crimson Gloves, and Obsidian Boots generate substantially lower USD revenue compared to Leather Boots, indicating a highly skewed revenue distribution.

- Several items, including Iron Boots, Half Plate, and Sandals, generate relatively high GEM revenue while exhibiting moderate or low USD revenue. This suggests that revenue contribution differs across currencies depending on the item.

- Some items show minimal or no USD revenue (e.g., Leather Armour and Leather Gloves), despite having non-zero GEM revenue or unit sales, indicating that these items are primarily purchased using in-game currency rather than real currency.

- For all listed items, the total number of units sold is equal to the number of purchase transactions, implying that purchases in the dataset predominantly consist of single-unit transactions.

Conclusion

The analysis shows that virtual item revenue is concentrated among a small number of high-performing items, with Leather Boots generating substantially more revenue than all other items. Revenue composition varies across items, with some items contributing more through USD purchases and others primarily through GEM transactions. Overall, the results highlight uneven revenue distribution across the in-game marketplace.

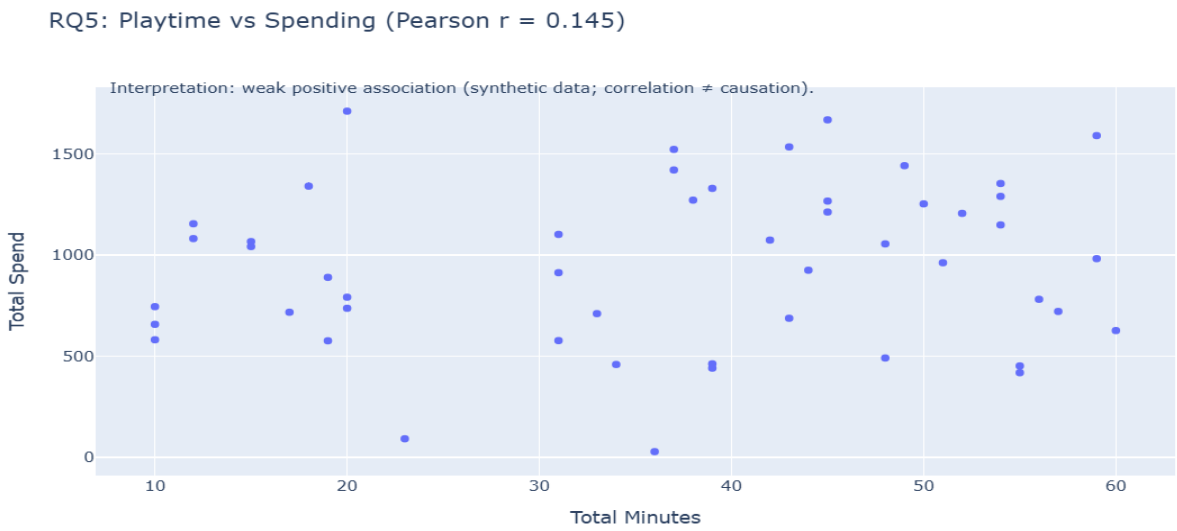
3.2.5 Research Question 5

Is there a correlation between playtime and total spending?

Data and Methodology

In this question, we aimed to find the correlation between playing time and money spent for skin purchases of players. We selected the total playing time from the GameSession table and the total expenditure from the SkinPurchase table. We created the scatterplot of total minutes and total expenditure in order to show how this relationship is.

Result



Interpretation

Analyzing the scatterplot, there is a weak positive relationship between total minutes and total spend. The correlation coefficient is equal to 0.145. As total duration in play gets increased, total money spent for purchases increase slightly. Correlation does not mean causation. As the values of total minutes gets close to each other, between 50-60, this positive relationship gets more apparent.

Conclusion

This analysis shows that the total minutes does not have the major impact on the total expending to items but it is a positive impact.

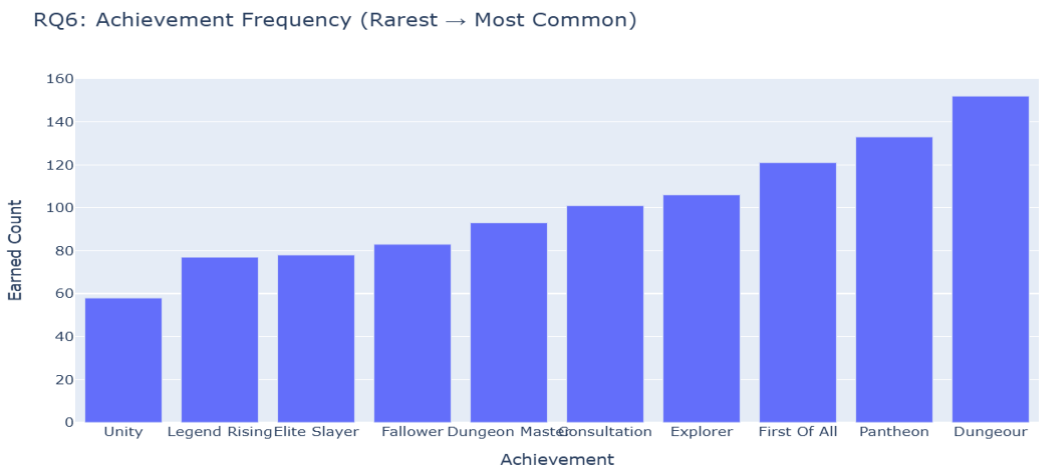
3.2.6 Research Question 6

Which achievements are the rarest and which are earned most frequently?

Data and Methodology

In this question, we focused on finding the rarest achievement and the most frequent one. We selected the count of players from Player table for each achievement obtained from Achievement. We created a bar graph to show the top 10 achievement earned the rarest in ascending order, from the rarest to the most common.

Results



Interpretation

By looking at the graph above, there is a regular and slow rising of the earned counts of the achievements generally. The earned counts vary between approximately 60 and 160. The rarest achievement earned is Unity with ~60 earnings and the most common one among them is Dungeon with ~150 earnings. It can be inferred that Unity, Legend Rising, Elite Slayer are some of the most difficult achievements and Pantheon, First of All and Dungeon are some of the easier achievements than the first three.

Conclusion

It shows that when we ordered the achievements in ascending order according to their earning, there is no rapid increase between the achievements. Therefore, there is no significant difficulty between the majority of achievement obtained.

3.2.7 Research Question 7

How do special in-game events affect player activity and spending?

Data Limitation and Analytical Approach

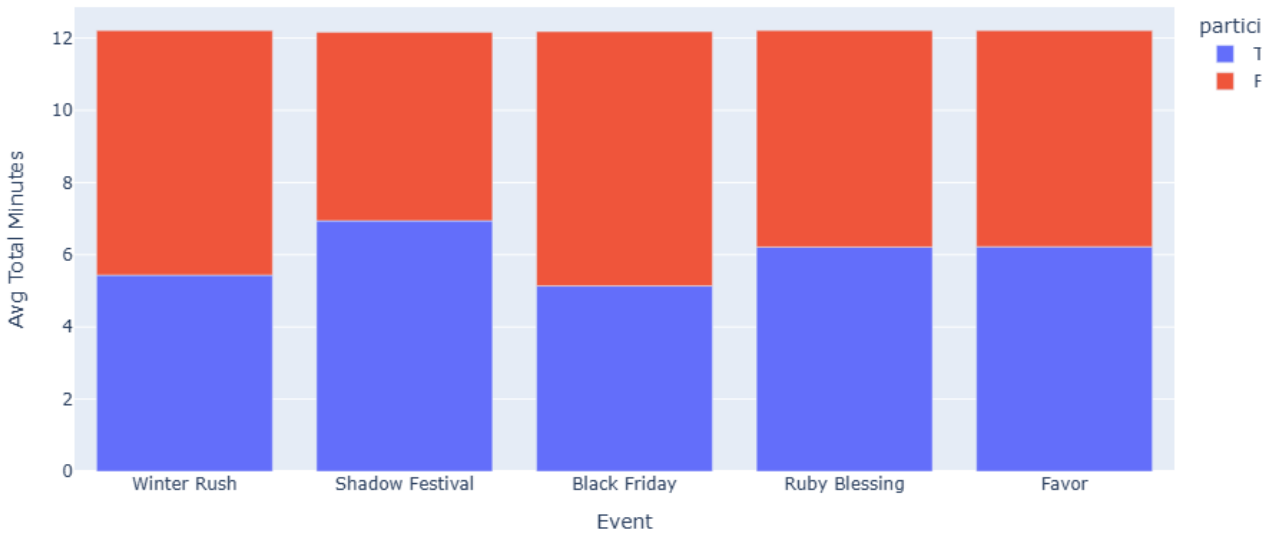
Ideally, the impact of in-game events on player behavior would be analyzed by comparing player activity and spending before, during, and after each event. However, the event-related tables in the database do not contain explicit event start and end time windows. Due to this limitation, a time-based analysis is not feasible.

To address this issue, we compare event participants and non-participants. For each event, players are grouped based on whether they participated, and their average total playtime and average total spending are calculated. This approach allows us to assess whether players who engage with events tend to be more active or spend more than those who do not.

SQL Query Design

To support this analysis, total playtime and total spending are first calculated at the player level using common table expressions (CTEs). Playtime is computed by summing active_time_in_mins from the game_session table, while spending is computed by summing quantity × unit_price from the skin_purchase table. These values are then joined with the event_participation and event tables.

Players are grouped by event and participation status, and average values are calculated for both metrics. This structure ensures that the comparison between participants and non-participants is consistent across all events.



Results: Player Activity (Average Playtime)

The visualization results reveal that the effect of event participation on player activity varies across different events:

- Winter Rush

Participants averaged 5.44 minutes, while non-participants averaged 6.77 minutes.

This suggests that Winter Rush participants were less active on average than players who did not participate.

- Shadow Festival

Participants averaged 6.94 minutes, compared to 5.22 minutes for non-participants.

This indicates a positive impact on player activity, with participants spending more time in the game.

- Black Friday

Participants averaged 5.14 minutes, whereas non-participants averaged 7.05 minutes.

Despite being a major promotional event, participants showed lower average playtime, suggesting that engagement may be more focused on purchases rather than extended gameplay.

- Ruby Blessing

Participants averaged 6.22 minutes, while non-participants averaged 6.00 minutes.

The difference is small but indicates slightly higher activity among participants.

- Favor

Participants averaged 6.22 minutes, compared to 5.99 minutes for non-participants.

This event shows a modest increase in activity for players who participated.

Overall, some events such as Shadow Festival appear to increase player engagement, while others show little or even negative impact on playtime.

Interpretation of Findings

These results suggest that not all in-game events affect player behavior in the same way. Events that are content-driven or challenge-based may encourage longer play sessions, while promotional or reward-focused events may attract participation without significantly increasing playtime.

The variation in outcomes highlights the importance of event design. Simply offering an event does not guarantee increased engagement; the structure and incentives of the event play a crucial role in shaping player behavior.

Conclusion

In conclusion, special in-game events have a mixed impact on player activity. Some events are associated with increased average playtime among participants, while others show no improvement or even lower engagement compared to non-participants. Due to the lack of event time windows, this analysis cannot establish causality, but it provides valuable insights into how different event types correlate with player behavior.

These findings suggest that future event designs should be carefully evaluated to ensure they meaningfully enhance player engagement, and additional data—such as event duration—would enable more precise analysis in future studies.

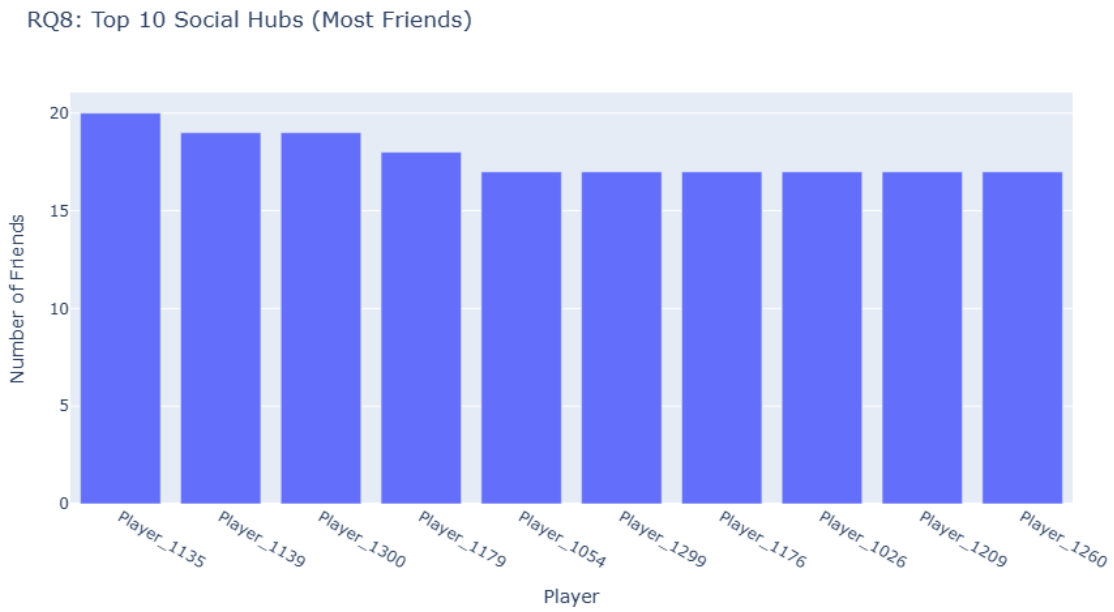
3.2.8 Research Question 8

Which players have the largest number of friendships (social hubs)?

Data and Methodology

Social interaction is an important factor in understanding player behavior in online games. To analyze social connectivity, we focused on the Friendship table in the database, which stores relationships between players. Each friendship involves two players, meaning a player can appear either as player_id or friend_id. In order to correctly measure how socially connected each player is, both columns were included in the analysis.

The SQL query aggregates friendship counts from both sides of the relationship using a UNION ALL operation. This ensures that all connections are counted without removing valid entries. The results are then grouped by player and ordered in descending order to identify the top ten players with the highest number of friends. This approach allows us to accurately capture each player's total number of social connections.



Results

The visualization created from the query results displays the ten players with the highest friendship counts in the game. The most connected player, Player_1135, has 20 friends. Several other players, including Player_1139 and Player_1300, follow closely with 19 friends. The remaining players in the top ten have between 17 and 18 friends.

The relatively narrow range of friendship counts suggests that social connectivity is not concentrated in a single player but rather distributed across multiple highly active individuals. This indicates a balanced and well-connected player community.

Interpretation and Discussion

Players with a high number of friends often act as central nodes within the game's social network. These players are more likely to interact frequently, participate in multiplayer activities, and encourage others to engage with the game. As a result, they may play an important role in increasing overall player activity, especially during special in-game events.

From a business and analytics perspective, socially active players are particularly valuable. They may influence the participation of others, contribute to longer play sessions, and indirectly impact in-game spending by motivating friends to log in and participate in time-limited events. This aligns closely with the project's research question, as social connectivity can amplify the effects of special in-game events on both player activity and spending.

Additionally, this analysis demonstrates how a well-designed relational database supports meaningful insights. The use of primary keys, foreign keys, and integrity constraints ensures that friendship data is consistent and valid. This reliability makes it possible to perform complex aggregations and joins without compromising the accuracy of results.

Conclusion

In conclusion, the analysis of player friendships reveals that social connectivity is an important characteristic of the game's player base. The presence of multiple highly connected players suggests a strong social structure that can enhance engagement and retention. These findings indicate that social features may strengthen the impact of special in-game events by encouraging collective participation and sustained activity.

Overall, this analysis highlights the value of combining a well-structured SQL database with analytical queries to uncover behavioral patterns. Understanding social connectivity provides a strong foundation for further exploration of how in-game events influence player activity and spending, supporting both analytical objectives and application-level decision making.

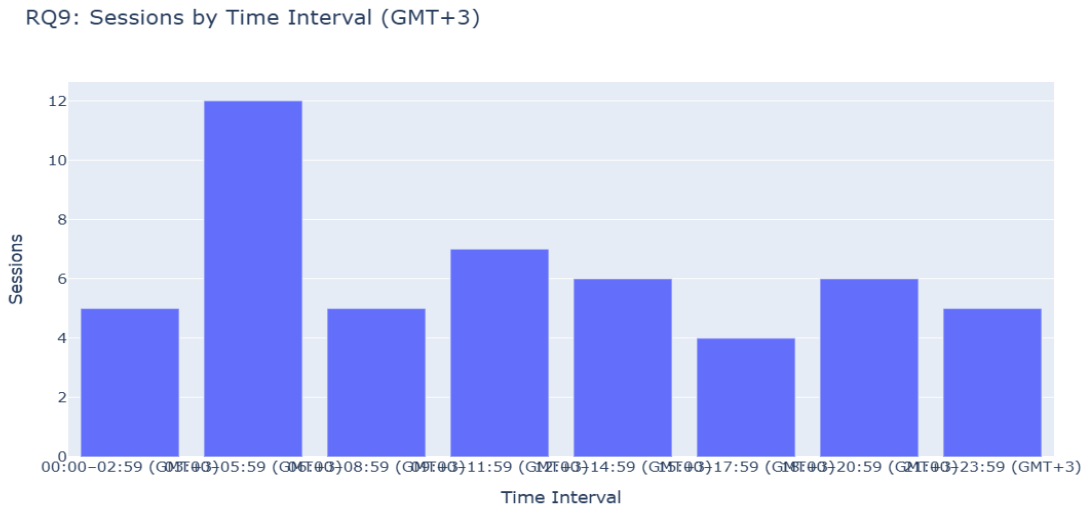
3.2.9 Research Question 9

What times of day have the highest number of active sessions?

Data and Methodology

In this question, we aimed to find the time having the most activity in a day. We selected the time intervals for 3 hours extracted from hour starting with the start time and the count of sessions for each interval. We adjusted the time zone according to the country we live in. We plotted a histogram showing the frequency of sessions for each hour intervals whose duration is 3 hours.

Results



Interpretation

There is no a regular distribution between the time intervals. The time interval having the most sessions is 03.00-05.59 with 12 sessions, and the time interval involving the least ones is 15.00-17.59 with 4 sessions. Players prefer to play more between midnight and noon, and the count of activities have lowest values in the afternoons.

Conclusion

It shows that players tend to play sessions more in late hours of a day and in the morning. The number of sessions shows the variability within a day.

3.2.10 Research Question 10

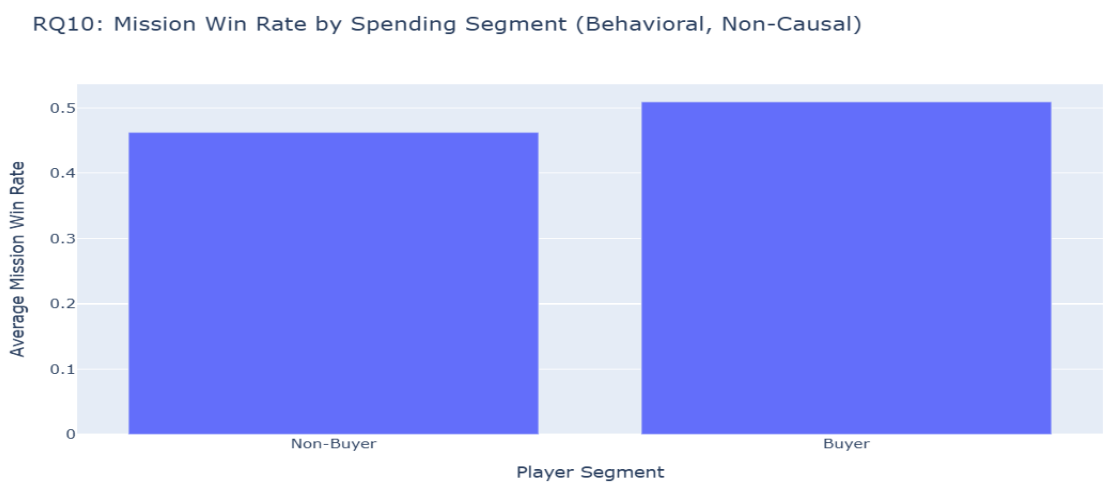
Do players who purchase items perform better in missions (win-rate comparison)?

Data and Methodology

We aimed to compare the mission win rates between players purchasing and players that do not have the habit of buying. We grouped players as buyer and non-buyer, and we found the players buying regarding whether their ID number is included in GameSession or not. We determined a case that demonstrates as 1 if a mission is won and otherwise is 0. We selected

the count of wins and failures from GameSession.mission_attempt and we proportioned the wins to the total mission attempts for each group. We constructed a bar chart showing the win rates for each group.

Results



Interpretation

The players that are purchasing items are more likely to contribute to the success of the missions than the players who do not prefer to buy. The win rate of buyers is ~%54 and the non-buyers' is around ~%46. Those proportions are close to each other and there is no considerable impact of buying items on obtaining success.

Conclusion

This analysis shows that spending money on items is one of the factors to raise the probability of resulting in success of a mission.

4. CONCLUSION

This report presents the design, implementation, and analysis of a relational database system developed for an online gaming environment. The study began with a conceptual analysis of the domain, identifying core entities, and defining relationships between them. Based on this model, a structured and normalized database schema was created to represent in-game activities and transactions while ensuring data integrity. The database was then implemented using SQL, providing a functional foundation for analysis.

Following the implementation, a set of research questions was formulated to examine different dimensions of the system within a unified analytical framework. These questions were analyzed using SQL-based queries to ensure consistency and comparability across results. The analysis shows that outcomes are not uniformly distributed across the system and vary across different dimensions, reflecting differences in performance and interaction

patterns among entities. Overall, the results demonstrate the value of aggregate analysis for identifying general patterns and trends. These trends can be obtained from https://yonelalper.github.io/gaming-analytics/dashboard.html

The relational structure and normalization of the database enabled consistent aggregation, comparison, and ranking of metrics, demonstrating how database design directly impacts how effectively data can be analyzed. In addition, the schema was designed to be flexible and extensible, allowing further analyses without major structural changes.

The analysis relies on a dataset created for this project to demonstrate database design and analytical techniques; therefore, the findings are illustrative rather than representative of real-world behavior. Overall, this study highlights how a well-designed relational database, combined with SQL-based analysis, provides a strong and scalable foundation for structured analytical exploration of complex systems such as online games.

References

Elmasri, R., & Navathe, S. B. (2016). *Fundamentals of database systems* (7th ed.). Pearson.

Silberschatz, A., Korth, H. F., & Sudarshan, S. (2019). *Database system concepts* (7th ed.). McGraw-Hill Education.

PostgreSQL Global Development Group. (n.d.). *PostgreSQL documentation*. https://www.postgresql.org/docs/